

MBA787M

STRATEGIC COST MANAGEMENT REPORT



GROUP - 8

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Subway Franchise Outlet – IIT Kanpur



Executive Summary & Business Profile

The outlet under study is a **Subway franchise unit located within the Indian Institute of Technology (IIT) Kanpur campus**. Operating as a **Quick Service Restaurant (QSR)**, the outlet specializes in **customizable submarine sandwiches (subs) and salads**, catering primarily to the food preferences of students, research scholars, faculty members, and campus staff. The campus location provides the outlet with a largely **captive customer base**, characterized by high repeat purchases and demand concentration around meal times.

Operating Model:

- **Operating hours:** 17 hours per day (08:00 AM – 01:00 AM).
- **Total staff strength:** 12 employees.
- **Shift structure:**
 - 8-hour shifts with 1-hour break
 - 4 employees per shift
 - 2 additional staff during peak hours (12 PM – 2 PM)
- **Seating capacity:** 28 seats.



Product List & Pricing

Best-Selling Products

- **Vegetarian:**
 - Paneer Tikka Sub
 - Aloo Patty
 - veg shami
 - Tandoori Paneer
 - cheesy mexican
 - veggie delite
- **Non-Vegetarian:**
 - Spicy Chicken Keema
 - Tandoori Chicken Tikka
 - American BBQ

Complementary Items

- Cookies
- Beverages

(Prices follow standard Subway campus pricing and are reflected in the outlet's average bill value.)

NEW HOT & CHEESY SIGNATURE SUBS		1.5X FILLING
VEG		6 INCH (15 CM)
Potato Chilli Cheese	₹239	
518kcal/238g		
Cheesy Mexican	₹259	
597kcal/281g		
Cheesy Paneer Tikka	₹289	
678kcal/299g		
NON VEG		
Chicken Keema Melt	₹239	
569kcal/233g		
Egg'n Chicken Melt	₹239	
525kcal/292g		
Cheesy American BBQ	₹259	
457kcal/230g		
Cheesy Chicken Tikka	₹289	
654kcal/299g		
Cheesy Lamb Pepperoni	₹299	
503kcal/227g		



Demand Forecast: The "Campus Calendar" Model

To forecast accurately, we cannot use a flat daily average. We have segmented the year into **3 Traffic Zones**.

Zone A: Peak Season (Fests & Events)

- **Trigger:** Cultural/Tech Fests (Antaragni, Techkriti) & Inter-IIT Sports.
- **Volume: 250 Orders/Day.**
- **Duration:** 26 Days (Oct, Mar, Dec, & Misc Holidays).
- **Characteristics:** High-ticket orders, bulk purchases, extended night operations.

Zone B: Regular Season (Semester)

- **Trigger:** Standard Academic Days.
- **Volume: 110 Orders/Day** (Weighted Avg of Weekday/Weekend).
- **Duration:** 219 Days.
- **Characteristics:** Consistent lunch/dinner rush, steady "Meal of the Day" sales.

Zone C: Lean Season (Vacations)

- **Trigger:** Summer Break (May-July) & Winter Break (Nov 28 - Jan 2).
- **Volume: 75 Orders/Day** (30% Drop).
- **Duration:** 120 Days (Excluding 8 days of Inter IIT which fall in Winter).
- **Characteristics:** Staff/Faculty residents only. Low footfall.



Annual Demand Estimation

Season	Orders/Day	Total Days	Total Annual Orders	% of Traffic
Peak	250	48	12,000	31%
Regular	110	194	21,340	56%
Lean	55	123	6,765	13%
TOTAL	~109 (Avg)	365	40,105	100%

3. Operations Analysis

☐ Procurement & Inventory (Hybrid Model)

Centralized Sourcing (15-Day Cycle)

- Frozen patties, sauces, cheese sourced from Delhi.
- Ensures standardization and bulk cost advantage.
- Risk: High holding cost during lean months.
- Mitigation: Orders reduced by ~60% during vacation period.
- Wastage maintained at only **2–3%**.

Local Sourcing (3-Day Cycle)

- Fresh vegetables procured locally.
- Follows **Just-in-Time (JIT)** approach.
- Ensures freshness and reduces spoilage.

☐ Purchase Process

- Outlet manager places orders based on:
 - Demand forecast (Peak/Regular/Lean zones)
 - Inventory levels
- Average purchase order: ~290 units.
- Stock checked at delivery and recorded daily.

❑ Inventory Management

- FIFO method followed strictly.
- Monthly raw material cost: ₹2.5–3 lakhs.
- Bread prepared in-house.
- Cold storage support:
 - 3 deep freezers & 2 chillers

❑ Service Delivery Model (Process vs Job Driven)

Back-End (Process Driven)

- Bread baking and vegetable preparation are continuous, standardized processes.
- Cost calculated per batch.

Front-End (Job Driven)

- Assembly line system.
- Each sub is customized.
- Flow: Order → Assembly → Billing → Delivery.

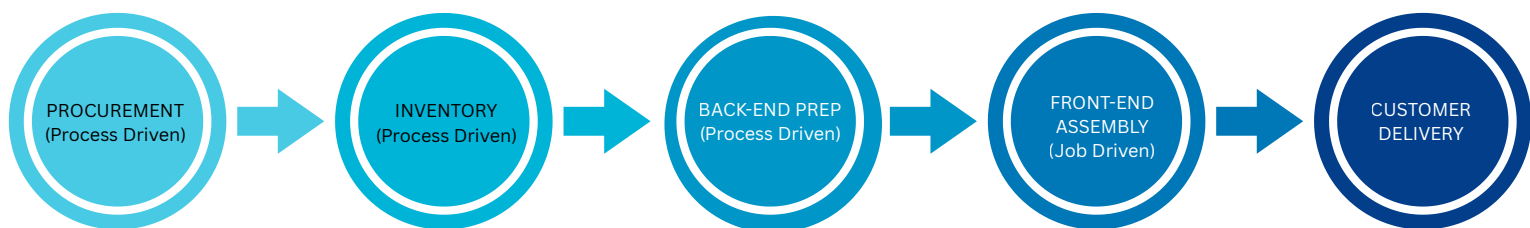
Since each sub is made according to individual preference, it follows **Job Costing**.

❑ Bottleneck & Improvement

- Bottleneck at **Sauce/Vegetable station** during peak fests (250 orders/day).
- Recommendation: Introduce **“Express Subs” (no customization)** during events.
- Expected impact: ~30% faster service during high demand.

❑ Capacity & Staffing

- 12 staff during semester.
- Reduced to 8 during vacations.
- Extra 2 staff during lunch peak (12–2 PM).
- Seating capacity: 28.



4. Costing Exercise (Dynamic Cost Structure)

The key to profitability is the **Variable Fixed Cost** strategy implemented below.

A. Fixed Costs (Monthly Burden)

Cost Head	Regular Season (Jan-Apr, Aug-Nov)
Rent	₹ 1,00,000
Staffing	₹ 1,74,000
Maintenance	₹ 10,000
TOTAL FIXED	₹ 2,84,000

B. Variable Costs (Per Unit)

- **Food Cost:** 32.0%
- **Royalty (8%) + Ad Fund (4.5%):** 12.5%
- **Packaging & Waste:** 4.0%
- **Total Variable Cost:** 48.5%
- **Contribution Margin: 51.5%** (Per ₹1 of sales, ₹0.515 covers Fixed Costs).

5. Breakeven & Profitability Analysis

Annual Summary

- **Total Revenue:** ₹ 1.16 Cr
- **Gross Contribution:** ₹ 59.9 Lakhs (51.5%)
- **Total Fixed Costs:** ₹ 43.1 Lakhs (After Lean Savings)
- **Net Profit:** ₹ 16.8 Lakhs
- **Net Margin:** 14.5%
- **From BEP table:**

- At 1844 units → Profit ≈ zero
- At 2000 units → Profit positive
- At 3000 units → Profit ≈ ₹1.78 lakh
- At 3090 units (observed month) → Profit ≈ ₹1.92 lakh
- Profit margin at operating level ≈ 20–30%

Monthly Break-Even Analysis

- **BEP Formula:** Fixed Cost / 0.515

Monthly_Financial_Perfor...															
Month	#	Days (Peak)	#	Days (Reg)	#	Days (Lean)	Revenue (₹)	#	Var Cost (46.87%)	Fixed Cost (Actual)	Net Profit (₹)	Net profit after initial cost	Remark(Assumptions)		
Nov		1		26		3	967,150		453,303	284,000	229,847	-438,153	Peak- Diwali, Lean- Endsem holidays		
Dec		8		0		23	1,080,250		506,313	284,000	289,937	-148,216	peak-Inter iit,lean-winter holiday of UG students		
Jan		1		28		2	1,009,200		473,012	284,000	252,188	103,972	peak-republic day,lean-winter holiday of UG students		
Feb		3		25		0	1,015,000		475,731	284,000	255,270	359,241	Peak- Valentine's week		
Mar		4		27		0	1,151,300		539,614	284,000	327,686	686,927	Peak-techkriti		
Apr		0		30		0	957,000		448,546	284,000	224,454	911,381			
May		0		0		31	674,250		316,021	284,000	74,229	985,610	Lean-Summer vacation		
Jun		0		0		30	652,500		305,827	284,000	62,673	1,048,283	Lean-Summer vacation		
Jul		0		0		31	674,250		316,021	284,000	74,229	1,122,512	Lean-Summer vacation		
Aug		1		30		0	1,029,500		482,527	284,000	262,973	1,385,486	Peak-Independence day		
Sep		0		30		0	957,000		448,546	284,000	224,454	1,609,940	NA		
Oct		8		23		0	1,313,700		615,731	284,000	413,969	2,023,908	Peak-Antaragni, Udghosh		
Total(Annual)		26		219		120	11,481,100		5,381,192	3,408,000	2,691,908				

Nov FC: Weighted avg (27 days Regular / 3 days Lean).

Dec FC: Weighted avg (23 days Lean / 8 days Peak). Even though Dec is a holiday month, the 8-day Inter IIT Event + reduced costs makes it **highly profitable**.

6. Top 7 Products Analysis (Pricing & Markup)

Product_Cost_Analysis													
Product Name	Category	Selling Price	Protein Cost	#	Bread Cost	Veggies Cost	Cheese Cost	Sauces Cost	Packaging	Total COGS	Contribution (₹)	Margin %	Markup
Cheesy Paneer Tikka	Veg	289	40		20	15	10	4.8	6	95.8	193.2	66.85%	201.67014
Aloo Patty	Veg	169	10		20	5	5	4.8	6	50.8	118.2	69.94%	232.67716
Veg Shami	Veg	199	10		20	20	5	4.8	6	65.8	133.2	66.93%	202.43161
Chicken Keema Melt	Non-Veg	239	30		20	10	5	4.8	6	75.8	163.2	68.28%	215.30343
Tandoori Chicken Tikka	Non-Veg	239	20		20	10	5	4.8	6	65.8	173.2	72.47%	263.22188
Cheesy American BBQ	Non-Veg	259	30		20	15	10	4.8	6	85.8	173.2	66.87%	201.86480
Cheesy Mexican	Non-Veg	259	20		20	20	10	4.8	6	80.8	178.2	68.80%	220.54455
Veggie Delite	Veg	159	10		20	20	5	4.8	6	65.8	93.2	58.62%	141.64133

Conclusion: All products exceed the manager's 20-30% target. The lowest margin is 45.5% (Chicken Tikka), leaving ample room for overheads.



Key Assumptions Summary

1. 6-inch format used.
2. No GST included in calculation.
3. Electricity partly treated variable.
4. Manager provided salary structure.
5. Average order value $\approx$ ₹290.
6. Working days considered 30 per month.
7. Inventory cycle:
 - Vegetables: 3 days (local vendor)
 - Sauces/Bread: 15 days (centralized supply)
8. Seating capacity: 28.
9. Shifts: 3 shifts, 12 staff total.
10. Profit margin target $\approx$ 30%.

Conclusion

The campus Subway outlet operates under a high contribution, moderate fixed cost QSR model.

Key findings:

- Break-even at approx. 1,844 units/month
- Average margin: 65%+ gross
- Operating profit margin: 20–30%
- Strong profitability in peak academic months
- Lean periods require demand stimulation

The outlet is financially viable and sustainable under current demand assumptions.

References:

1. We have mention all the assumption in [excel sheet](#) .
2. Data given by Manager.
3. Rates of some items taken from hall 2 mess.[rates](#)